

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 3 MODULE, AUTUMN SEMESTER 2011-2012

COMPUTER VISION

Time allowed TWO Hours

Candidates must NOT start writing their answers until told to do so

Answer THREE out of FIVE questions

No calculators are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

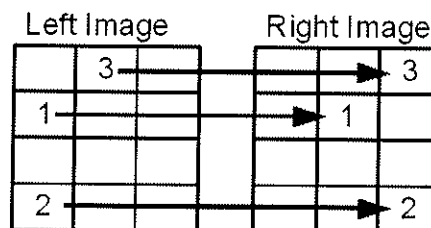
1. Answer all of the following sub-questions pertaining to Stereo Correspondence (25 marks):

(a) Explain what kinds of situations lead to sparse and dense disparity maps.

[4 marks]

(b) Consider the sparse correspondence map below. Note that features are denoted by numbers (i.e. 1, 2 and 3), correspondences are denoted by arrows and empty cells denote unmatched coordinates. Draw a plausible dense disparity map from the sparse disparity map. Note that the sparse disparity map can be easily generated from the correspondences below. Discuss the strategies that you adopted when generating the dense disparity map.

[7 marks]



(c) Describe the FOUR main steps behind the computation of depth maps.

[6 marks]

(d) Algorithms that compute disparity maps often produce certain artifacts that require cleaning up. Discuss the nature and causes of these artifacts, and discuss possible solutions for cleaning them up.

[4 marks]

(e) Discuss two important assumptions that help solve the correspondence problem and contribute towards the computation of disparity maps.

[4 marks]

2. Answer all of the following sub-questions pertaining to Segmentation (25 marks):

- (a) Consider a segmentation algorithm that forms clusters of pixels based on distinctive peaks in the intensity histogram of some image. Is the algorithm a region based segmentation algorithm? Justify your answer.

[4 marks]

- (b) Consider a crowded environment (e.g. rock concert). How might you build a simple system for detecting unruly crowd behaviour based almost entirely on segmentation?

[5 marks]

- (c) Describe the region growing approach to segmentation. In what way(s) can seed initialization be a problem?

[4 marks]

- (d) Work through TWO different k-Means clustering examples. In both cases attempt to cluster the set of numbers $\{2,4,8,9\}$. In the first case use $k=2$ and initialize the means to 2 and 4. In the second case use $k=3$ and initialize the means to 2, 4 and 6. Note that the calculations required are simple enough to preclude the need for a calculator.

[12 marks]

3. Answer all of the following sub-questions pertaining to Motion Analysis and Optic Flow (25 marks):
- (a) Consider a simple motion detection algorithm based on frame differences. Use this approach in order to explain the problems of camouflage and occlusion. Discuss one possible solution for each problem.
[8 marks]
 - (b) Explain why and how image pyramids are used in the computation of optical flow.
[8 marks]
 - (c) It is a well-known issue that optical flow at the boundaries of objects is often inconsistent and ambiguous. Explain the concepts of forward/backward flows and anisotropic diffusion and how they can help address this issue.
[6 marks]
 - (d) Give three examples of where the optical flow computed from a sequence of frames does not correspond to the motion field of the respective scene.
[3 marks]

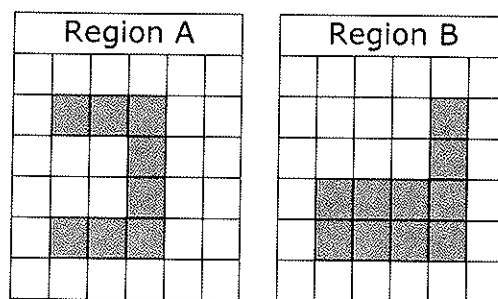
4. Answer all of the following sub-questions pertaining to Evaluation and Tracking (25 marks):

- (a) The evaluation of segmentation algorithms is not always straightforward. Since segmentation algorithms are often early processing components in larger systems involved in higher level visual functions it is often useful to evaluate segmentation algorithms in terms of their relative performance in the overall function of the system they are embedded in. Give an example of such a system and explain how you might evaluate different segmentation algorithms based on that system.

[6 marks]

- (b) Compute the earth mover distance between regions A and B – $\text{dist}(A,B)$ - and between regions B and A – $\text{dist}(B,A)$ - below.

[5 marks]



- (c) Explain how the earth mover distance can be used in the evaluation of segmentation algorithms.

[6 marks]

- (d) What is the aperture problem and how does it relate to the problem of tracking?

[3 marks]

- (e) In the context of tracking, what is the template update problem? Briefly suggest a simple solution to address this problem.

[5 marks]

5. Answer all of the following sub-questions pertaining to Hidden Markov Models (25 marks):
- (a) Describe the three main functions used by Hidden Markov Models. Use a real-world example in order to complete your description.
[5 marks]
 - (b) Explain in detail a simple (and computationally expensive) method based on Hidden Markov Models for computing the probability of a sequence of observations.
[8 marks]
 - (c) Consider the context of a train station, where typical behaviours involve queuing, buying tickets, walking to/from a train, and so on. Describe how you might use Hidden Markov Models in order to detect any abnormal behaviour.
[5 marks]
 - (d) Consider the problem of hand gesture recognition, where gestures are distinguished primarily by motion characteristics rather than (hand) shape configurations. Describe a solution to the problem based on Hidden Markov Models. Don't forget to include in your description the early processing stages (e.g. feature extraction) of the solution.
[7 marks]